Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/12/2009
PI Name:	Bateman, Ted A. Ph.D.		
Project Title:	Examination of Anti-Resorptive and Anabolic Treatments/Stimuli on Unloading Induced Osteoporosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Chapel Hill	**State:**	NC
Zip Code:	27599	**Congressional District:**	4
Comments:	Previous affiliation was Clemson University; PI moved to UNC in fall 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed end date to 9/30/2008 (from 7/31/2008) per K. Major/NSBRI (6/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Simske, Steven (University of Colorado at Boulder) Norrdin, Robert (Colorado State University)		
Grant/Contract No.:	NCC 9-58-BL00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Studies examining the calcium balance during extended spaceflight on the Russian Space Station Mir and the International Space Station suggest that 1) both bone resorption and formation are altered to adversely affect bone density, and 2) the time required to return bone mass to pre-flight levels may be considerably longer than the period of microgravity exposure. Countermeasures to prevent the loss of bone mass, and therapies to more quickly replace bone post-flight, are desirable to not only prevent fractures during long-duration missions, but to also prevent microgravity-associated bone loss from exacerbating naturally occurring osteoporosis. We developed a treatment regimen that would prevent in-flight bone loss and accelerate the recovery of bone health post-flight. Both bone mass and turnover rates are important for evaluating successful recovery. The significance of these factors is illustrated by our demonstrating that low doses of anti-resorptive agents can prevent in-flight loss of bone mass and allow normal bone formation rates. Through a series of iterative studies in mice which were administered either osteoprotegerin (OPG) or zoledronate, doses of 500 µg/kg for OPG and 45 µg/kg, respectively, were determined to be the minimum, efficacious comparable doses in that they increased bone mass, while not causing the inhibition of bone formation that generally accompanies antiresorptive therapies. These doses of OPG and zoledronate were then tested for their ability to prevent disuse osteoporosis with the hindlimb suspension model in mice. Both antiresorptive therapies, at these lower doses, maintained bone mass, strength and structural properties at the level of fully loaded, placebo treated control mice. This study demonstrates the potential benefits of minimum dosing. The final year of the grant examined the sequential administration of bisphosphonates and anabolic PTH therapy. A mouse model was used to determine if anabolic treatment after antiresorptive therapy and hindlimb suspension can reverse the inhibition of bone formation that both unloading and antiresorptive therapy cause. The primary metric for this examination is an analysis of serum for markers of bone formation and resorption indicating increased bone turnover after treatment with zoledronate, with additional study quantitative histomorphometry for cortical bone formation rates. The duration of suppressed bone turnover was marginally reversed with PTH administration, however, the dose was higher than what has been reported in the literature, possibly because cortical bone has slower turnover rates than trabecular bone. A dose of 50ug/kg/day did increase systemic markers for bone turnover and bone formation rates, accelerating the recovery of these markers after zoledronate treatment by several days. We have also modified this aim to include a clinical examination of sequential antiresorptive and anabolic therapies in women receiving treatment for osteoporosis. Preliminary data collected in 105 patients suggest that prior bisphosphonate treatment inhibits the anabolic action of PTH. Our retrospective chart review was expanded to include up to 600 patients, allowing a more definitive determination of the efficacy of sequential therapy and potentially discriminate differences between bisphosphonates. We also added a study in the final year of this project examining the protein macrophage-colony stimulating factor (M-CSF) for its potential as a novel anabolic therapy for bone. Our studying M-CSF has been an ongoing, but low priority component of our lab's work. Our recent emphasis on radiation-induced bone loss was motivation to perform a simple, but comprehensive study with M-CSF; the contribution of radiation exposure to increased lifetime cancer risk is of concern in astronauts on long-duration missions as they will be exposed to moderate cumulative doses of proton and heavy ion radiation from solar and cosmic sources. In this context, the administration of PTH, or any anabolic agent to improve bone structure and strength may not be appropriate as it may synergistically act to further increase cancer risk. Though the contribution of PTH therapy to cancer risk has not been identified in humans, and the data are limited in animals, the use of any anabolic agent should be considered carefully. For this study, a group of mice was administered a relatively high dose of 5 mg/kg/day of huM-CSF for 21 days. In addition to a rapid increase in body mass, M-CSF increased serum bone turnover markers for both bone resorption and bone formation. Microcomputed tomography revealed an anabolic effect on tibial trabecular bone, with higher bone volume fraction (+35%), connectivity density (+79%), and number (+18%), as well as lower trabecular separation (-18%). M-CSF had no significant effect on cortical bone mineral content, geometry, or strength. There was no change in quantitative histomorphometry parameters of femoral cortical bone. Similar to our observations from radiation-induced bone loss, the "anabolic" effects of M-CSF appear to be specific to trabecular bone. In summary, during the four year duration of this study, we demonstrated a proof of concept comparison of two antiresorptive bone agents for minimum, but efficacious doses to prevent bone loss from disuse while minimizing the long-term suppression of bone turnover. We have also performed a comprehensive examination of human patients receiving PTH after treatment with a bisphosphonate, and are finalizing study of an expanded group of subjects. A similar regimen development in mice was initiated, and is still ongoing with the doses of PTH higher than anticipated. Finally, the circulating protein M-CSF was preliminarily identified as a potential therapy to increase bone turnover.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The human skeletal system consists of a dynamic balance between solid, calcified bone and free Ca2+ in the blood. Two processes, resorption and formation, consistently turnover bone and contribute to "bone remodeling". An imbalance in this system, such as in post-menopausal women or during spaceflight, can result in a shift favoring bone resorption. The result of this imbalance is low bone density, bone mass and an increase risk of fracture. Research in the area of spaceflight induced bone loss has implications for the causes of osteoporosis on earth, and in particular the development of effective therapies to prevent and reverse bone loss. Indeed, studies on the International Space Station reveal that bone loss occurs at a rate much faster than of post-menopausal woman. With over 1.5 million fractures occurring per year due to the proliferation of an aging population and the expansion of space exploration to unprecedented new lengths, the need for the development of a variety of advanced, effective therapies has never been greater. Both astronauts as a population and spaceflight as a cause are significantly different than the population and pathology current osteoporosis therapies were designed to treat. It should not be presumed that doses and therapies intended for older women are appropriate and efficacious for astronauts on exploratory missions. Optimizing treatment regimens for populations, like astronauts, may eventually be expanded to modifying regimens to individuals, depending on specific physiological deficits and response.

Task Progress:	Aim 1 (completed): In the first year of this grant we developed a treatment regimen that would prevent in-flight bone loss and accelerate the recovery of bone health post-flight. Through a series of iterative studies in mice administered either osteoprotegerin (OPG) or zoledronate, doses of 500 µg/kg for OPG and 45 µg/kg, respectively, were determined to be the minimum, efficacious comparable doses: they increased bone mass, while not causing the inhibition of bone formation that generally accompanies antiresorptive therapies. Aim 2 (completed): These doses of OPG and zoledronate were then tested for their ability to prevent disuse osteoporosis with the hindlimb suspension model in mice. Both antiresorptive therapies, at these lower doses, maintained bone mass, strength and structural properties at the level of fully loaded, placebo treated control mice. This study demonstrates the potential efficacy of minimum dosing. The most significant finding from hindlimb suspension study is the potential synergistic inhibition of bone formation caused by the combination of antiresorptive therapies and disuse. Though both doses of OPG and zoledronate caused no inhibition of bone formation in normally loaded control mice, the antiresorptive therapies caused an inhibition of bone formation of 42-44% in the hindlimb unloaded mice in addition to the 45% inhibition of bone formation caused by hindlimb suspension with no drug treatment. The results from the first two Aims were published in the "Journal of Applied Physiology" in 2008. Aim 3: The remaining stages of this project are examining the sequential administration of bisphosphonates and anabolic PTH therapy. This is being done to determine if anabolic treatment after disuse can reverse the inhibition of bone formation that both unloading and antiresorptive therapy cause. The duration of suppressed bone turnover was marginally reversed with PTH administration, however, the dose was higher than what has been reported in the literature. We have also expanded this aim to include a clinical examination of sequential antiresorptive and anabolic therapies in women receiving treatment for osteoporosis. Preliminary data collected in 105 patients suggest that prior bisphosphonate treatment inhibits the anabolic action of PTH. We added a study in the final year of this project examining the protein macrophage-colony stimulating factor (M-CSF) for its potential as a novel anabolic therapy for bone. M-CSF increased serum bone turnover markers for both bone resorption and bone formation. Microcomputed tomography revealed an anabolic effect on tibial trabecular bone, with higher bone volume fraction (+35%), connectivity density (+79%), and number (+18%), as well as lower trabecular separation (-18%). M-CSF had no significant effect on cortical bone mineral content, geometry, or strength.
Bibliography Type:	Description: (Last Updated: 11/12/2020)
Abstracts for Journals and Proceedings	Lloyd SA, Bateman TA. "Disuse exacerbates suppression of bone formation with administration of low-dose antiresorptive drugs." American Society for Gravitational and Space Biology 23rd Annual Meeting, Moffett Field, CA, October 25-28, 2007. Gravitational and Space Biology 2007 Oct;21(1):18. , Oct-2007
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Articles in Peer-reviewed Journals	Allen DL, Bandstra ER, Harrison BC, Thorng S, Stodieck LS, Kosteniuk PJ, Morony S, Lacey DL, Hammond TG, Leinwand LL, Argraves WS, Bateman TA, Barth JL. "Effects of spaceflight on murine skeletal muscle gene expression." J Appl Physiol. 2009 Feb;106(2):582-95. Epub 2008 Dec 12. PMID: 19074574 , Feb-2009
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Articles in Peer-reviewed Journals	Lloyd SA, Yuan YY, Simske SJ, Riffle SE, Ferguson VL, Bateman TA. "Administration of high-dose macrophage colony-stimulating factor increases bone turnover and trabecular volume fraction." Journal of Bone and Mineral Metabolism, Submitted, November 2008. , Nov-2008
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Awards	Tirado L. "2008 ACC Undergraduate Research Conference (one of ten Clemson representatives), Florida State University, Tallahassee, Florida, April 19, 2008." Apr-2008
Awards	Tirado L. "Travel Award from Society for Advancement of Chicanos and Native Americans in Science (SACNAS) for the International Polar Year: Global Change in Our Communities Conference, Salt Lake City, Utah, October 09-12, 2008." Oct-2008
Awards	Lemus M. "Undergraduate Research Fellowship from the South Carolina Space Grant Consortium, April 2008." Apr-2008

Awards	Riffle S. "2008 ACC Undergraduate Research Conference (one of ten Clemson representatives), Florida State University, Tallahassee, Florida, April 19, 2008." Apr-2008
Awards	Bateman TA. "Halstead Young Investigator's Award, American Society for Gravitational and Space Biology (ASGSB), November 2007." Nov-2007
Awards	Anderson E. "Best Undergraduate Research Presentation and Travel Awards from the American Society for Gravitational and Space Biology, November 2007." Nov-2007
Awards	Bowman L. "Graduate Student Research Program Fellowship from the South Carolina Space Grant Consortium, April 2008." Apr-2008
Books/Book Chapters	Bateman TA, Bandstra ER, Willey JS, Lloyd SA, Yuan YY, Bourne J. "How animal models inform the debate." in "Bone loss during spaceflight : etiology, countermeasures, and implications for bone health on Earth." Ed. P.R. Cavanagh, A.J. Rice. Cleveland : Cleveland Clinic Press, c2007. p. 189-199., Dec-2007